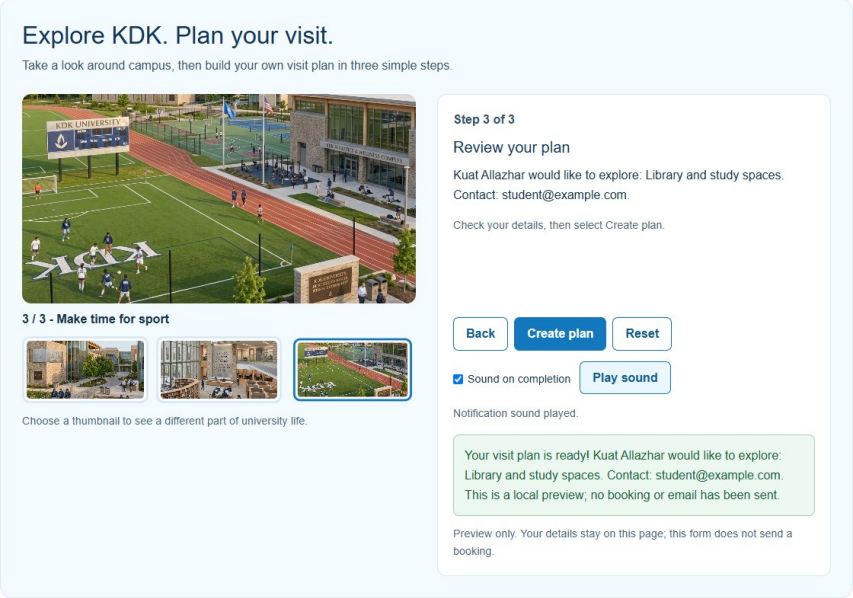
WEB TECHNOLOGIES 1

Assignment 5

Advanced JavaScript and DOM Manipulation

Submitted by	Kuat Allazhar
Team / group	theboys / SE-2515
Team members	Kuat Allazhar, Dautbekov Yerkhan, Khafiz Akylzhan
My pages	Home (index.html) and About Us (about.html)
Date	26 October 2026

Objective. Continue the KDK University team website with an interactive campus gallery and a three-step visit planner. The same features are available on both of my pages. Other members' pages and scripts are preserved.



Home: campus gallery, review step and successful plan preview.

Live website: https://wwezzy.github.io/kdk-university-project/

About: https://wwezzy.github.io/kdk-university-project/about.html

Repository: github.com/wwezzy/kdk-university-project

The selected requirements use plain JavaScript. For the alternative options, the implementation uses a thumbnail gallery, direct style changes and a multi-step form.

1. DOM manipulation and styling

1. Select the main image, caption and thumbnail container by their unique IDs. Store three campus photos in an array of objects.

2. Create thumbnail buttons with a for loop. A click calls showCampusPhoto with the selected index.

3. Change the main image's src and alt attributes, then update the caption with textContent.

4. Select the thumbnail group with querySelectorAll. Use forEach to update aria-pressed, the border color and transform of every button.

js/allazhar-assignment5.js

```
30  function showCampusPhoto(index) {
31    const photo = campusPhotos[index];
32    mainImage.setAttribute("src", photo.src);
33    mainImage.setAttribute("alt", photo.alt);
34    photoCaption.textContent = (index + 1) + " / " + campusPhotos.length + " - " + photo.title;
35    const buttons = thumbnails.querySelectorAll("button");
36    // forEach is a higher-order function: it receives a callback.
37    buttons.forEach(function (button, buttonIndex) {
38      const isSelected = buttonIndex === index;
39      button.setAttribute("aria-pressed", String(isSelected));
40      button.style.borderColor = isSelected ? "#1378bd" : "#d9ebf7";
41      button.style.transform = isSelected ? "scale(0.94)" : "scale(1)";
42    });
43  }
44  showCampusPhoto(0);
```



3 / 3 - Make time for sport





Choose a thumbnail to see a different part of university life.

Result: clicking Sports selects the third photo and changes the caption to “3 / 3”. The selected thumbnail has a blue border and a slightly smaller scale.

2. Events: keyboard navigation

1. Select all six links in the main navigation. Attach a keydown callback to each link.

2. Use switch(event.key) to handle Right/Down, Left/Up, Home and End. Modulo wraps focus from the last item to the first, and back.

3. Cancel the arrow key's default scrolling and call focus on the next link. Return for other keys so Tab and Enter retain their normal behavior.

js/allazhar-assignment5.js

```
51        switch (event.key) {
52          case "ArrowRight":
53          case "ArrowDown":
54            nextIndex = (index + 1) % menuLinks.length;
55            break;
56          case "ArrowLeft":
57          case "ArrowUp":
58            nextIndex = (index - 1 + menuLinks.length) % menuLinks.length;
59            break;
60          case "Home":
61            nextIndex = 0;
62            break;
63          case "End":
64            nextIndex = menuLinks.length - 1;
65            break;
66          default:
67            return;
68        }
69        event.preventDefault();
70        menuLinks[nextIndex].focus();
71      });
72    });
73
74    // 3. This object stores the form state and has its own method.
```



Result: Home is focused and has a visible outline. Arrow keys move this outline through the links. On mobile, open the Bootstrap menu first, then use the same keys.

Button events: thumbnails change photos; Next and Back change form steps; Reset clears the form; Play sound starts the notification. All handlers are registered with addEventListener.

2. Events: multi-step form

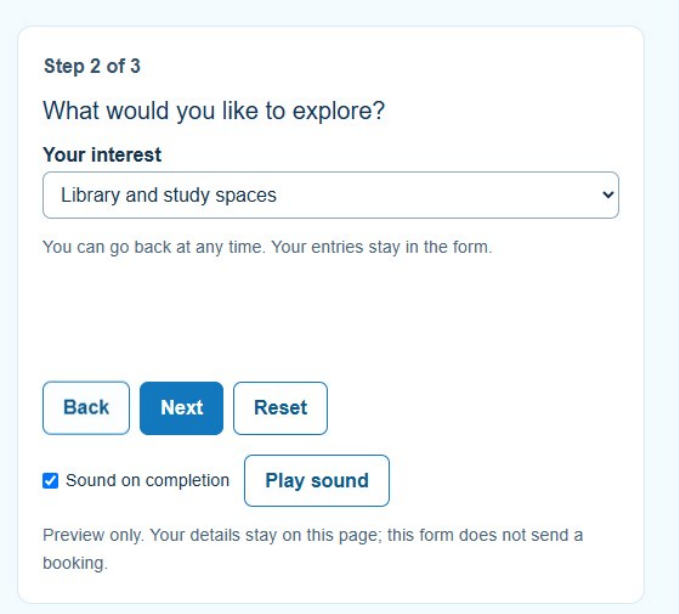
1. Read and validate name and email on step 1. Show an error and focus the relevant field when input is invalid.

2. Next and Back callbacks call showVisitStep. The hidden property changes which step is visible. The existing input elements stay in the DOM, so their values are preserved.

3. Choose an interest on step 2 and review the plan on step 3. Submit uses preventDefault and displays a local summary. Reset clears the form, saved values and messages.

js/allazhar-assignment5.js

```
104   function showVisitStep(stepIndex) {
105     currentStep = stepIndex;
106     steps.forEach(function (step, index) {
107       step.hidden = index !== currentStep;
108     });
109     backButton.hidden = currentStep === 0;
110     nextButton.hidden = currentStep === steps.length - 1;
111     finishButton.hidden = currentStep !== steps.length - 1;
112     progress.textContent = "Step " + (currentStep + 1) + " of " + steps.length;
113     errorMessage.textContent = "";
114   }
    ...
133   nextButton.addEventListener("click", function () {
134     if (validateVisitStep()) {
135       result.hidden = true;
136       showVisitStep(currentStep + 1);
137       steps[currentStep].querySelector("input, select, h3").focus();
138     }
139   });
```



About page, step 2: the user selects Library and study spaces. Going Back retains the name and email; returning to this step retains the interest. This is the assignment's multi-step alternative to a server POST form.

3. Objects, arrays and functions

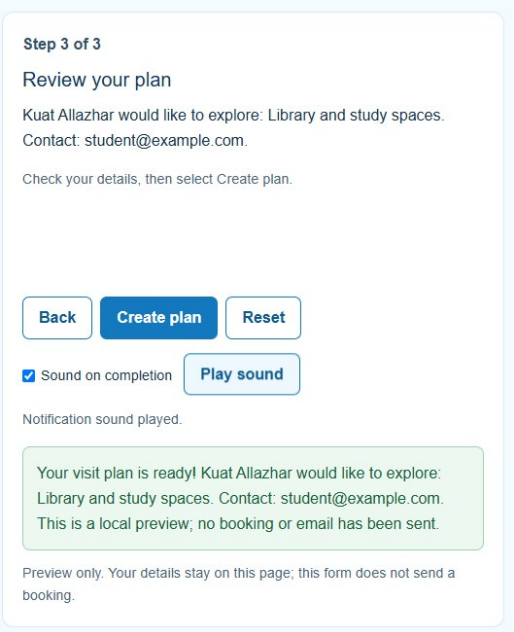
1. Use campusPhotos as a collection of three photo objects. Each object has src, title and alt properties. The for loop builds the thumbnail controls.

2. Store name, email and interest in visitPlan. Its getSummary method uses this to read the current object properties and returns a string.

3. Use forEach with callbacks for thumbnails, menu links and steps. This is a higher-order function because it takes a function as an argument.

js/allazhar-assignment5.js

```
75    const visitPlan = {
76      name: "",
77      email: "",
78      interest: "Campus tour",
79      getSummary: function () {
80        return this.name + " would like to explore: " + this.interest + ". Contact: " + this.email +
    ".";
81      }
82    };
83    const visitForm = document.getElementById("a5VisitForm");
    ...
15    for (let index = 0; index < campusPhotos.length; index++) {
16      const button = document.createElement("button");
17      const image = document.createElement("img");
18      button.type = "button";
19      button.className = "a5-thumbnail";
20      button.setAttribute("aria-label", campusPhotos[index].title);
21      image.src = campusPhotos[index].src;
22      image.alt = "";
23      button.appendChild(image);
24      thumbnails.appendChild(button);
25      button.addEventListener("click", function () {
26        showCampusPhoto(index);
27      });
```



Result on About: the object's current values appear in the review and success message. textContent treats the output as text instead of inserting user input as HTML.

3. Sound and event animations

1. Load one local WAV file with new Audio and set the volume to 25%. Play it after an explicit click, or when creating a plan with sound enabled.

2. Before each play, pause the existing sound and reset currentTime to zero. Reusing one object prevents overlapping copies. Handle a failed play promise with a readable message.

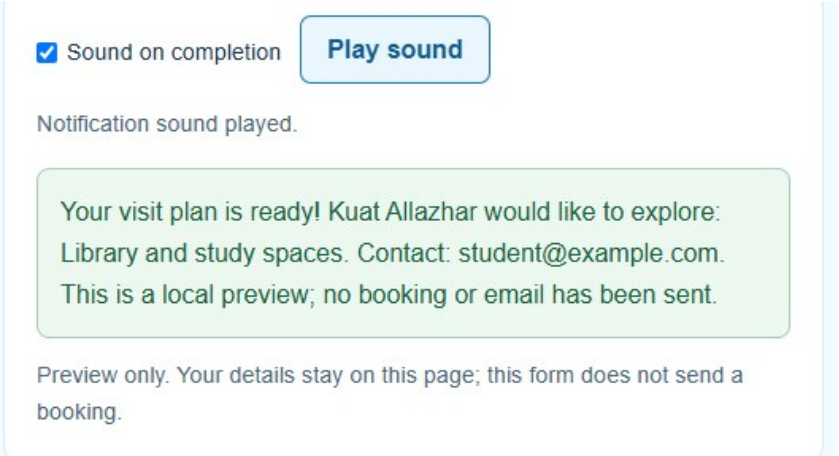
3. Animate selected thumbnail styles with a CSS transition. Show form steps and the result with opacity and translateY keyframes. Respect prefers-reduced-motion.

js/allazhar-assignment5.js

```
148   const notificationSound = new Audio("sounds/kdk-notification.wav");
149   notificationSound.volume = 0.25;
150   const soundMessage = document.getElementById("a5SoundMessage");
151   function playNotification() {
152     notificationSound.pause();
153     notificationSound.currentTime = 0;
154     notificationSound.play().then(function () {
155       soundMessage.textContent = "Notification sound played.";
156     }).catch(function () {
157       soundMessage.textContent = "Sound is unavailable. You can still use the visit planner.";
158     });
159   }
160   document.getElementById("a5PlaySound").addEventListener("click", playNotification);
161
```

css/allazhar-assignment5.css

```
88  @keyframes a5StepEnter {
89    from { opacity: 0; transform: translateY(10px); }
90    to { opacity: 1; transform: translateY(0); }
91  }
```



Result: “Notification sound played” appears above the animated success panel. Reset also stops the sound. Disabling Sound on completion keeps plan creation silent.

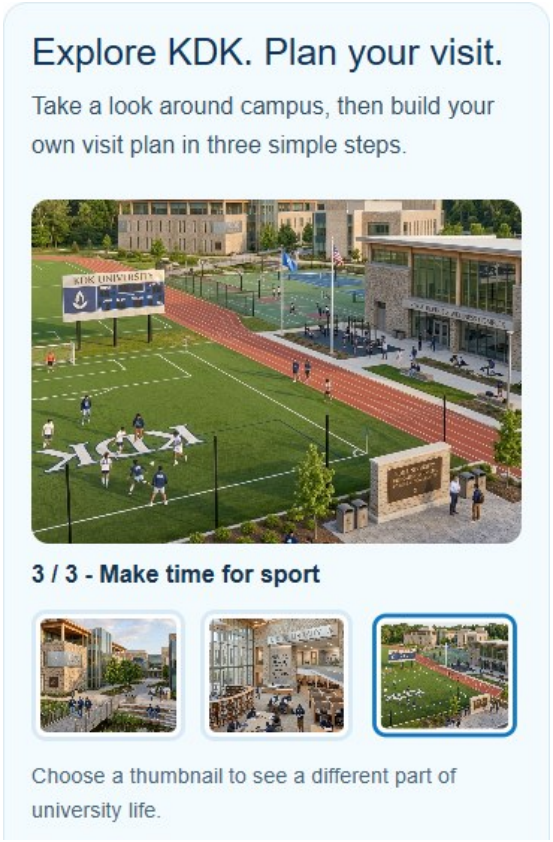
4-5. Structure and user experience

HTML contains the gallery and form. A separate CSS file contains the responsive layout and animation. A separate JavaScript file contains the behavior. Both pages load these files after the existing styles and script; no inline event attributes or style attributes were added to the HTML.

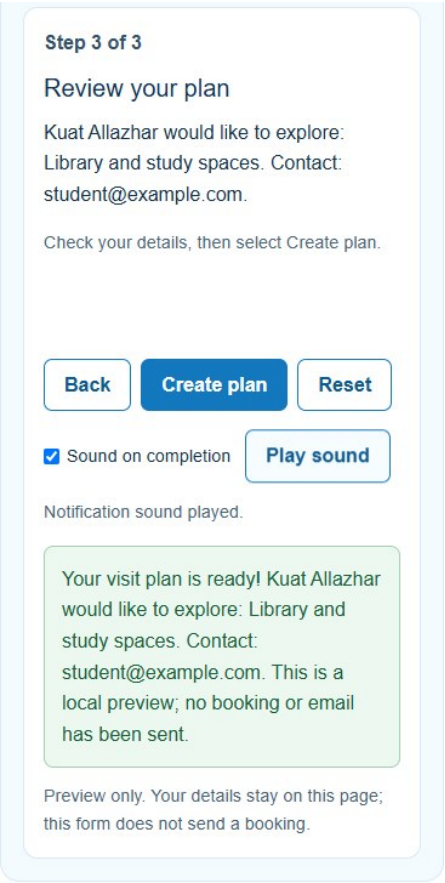
index.html

```
13      <link rel="stylesheet" href="css/allazhar.css">
14      <link rel="stylesheet" href="css/allazhar-assignment5.css">
    ...
348     <script src="js/allazhar.js"></script>
349     <script src="js/allazhar-assignment5.js"></script>
```

Theme and usability. Campus photos and a visit plan fit a university website. Named helper functions avoid repeated logic. Visible focus, input labels, descriptive photo text, progress, validation and Reset make the interface easier to use. A one-column layout fits a phone.



Home: mobile gallery.



About: mobile review and result.

The form clearly labels its result as a local preview. It does not send a real visit booking or email. The plan is intentionally kept only in page memory.

Deployment and final checks

The updated source was committed to the shared main branch and deployed using GitHub Pages. The project ZIP contains the complete website, including the other members' existing pages and assets. This report documents my Home and About contribution.

Source commit: e52fb64b8b774ccdfedfc8c6edf15603a39aedb3

Requirement	Implemented evidence
DOM / attributes / styles	Gallery src and alt; textContent captions; selected borders and transforms.
Events / callbacks / switch	Click listeners, keydown navigation, switch(event.key), Next and Back callbacks.
Objects / arrays / loops / HOF	visitPlan.getSummary, campusPhotos, for loop, forEach callbacks.
Sound / animation	One reusable local Audio object; transitions and step/success keyframes.
Structure / UX	External files; namespaced component styles; labels, focus, mobile layout and reduced motion.

Verification. The published site passed eight combinations: Home and About, Chrome and Edge, at desktop width 1280 px and mobile width 390 px. The screenshots in this report show the published website, with region crops used for detail.

Checks covered all thumbnails, arrow-key wrapping and Home/End, invalid input, Enter behavior, retained input on Back/Next, review text, Reset, sound on/off and reuse, responsive width, and reduced-motion styling. Existing FAQ, popup, registration errors, background button and date/time were also checked. No page JavaScript errors were recorded.

Challenges and reflection

Shared styles previously affected my FAQ, so the new component uses its own a5-prefixed classes and its own stylesheet. The old FAQ fix stays in place. Form steps keep the same input elements, which avoids losing entered data. A reusable Audio object prevents sound overlap.

This version demonstrates the selected assignment options with a small number of features. The visit planner has no backend, so it honestly shows a local preview instead of claiming a real booking. The next step is to explain each function and practise changing the array, switch cases, validation, sound volume and CSS animation during the defense.

Learning materials

Frontend Assignment 5; course JavaScript, DOM and Intermediate JavaScript slides; WT Week 8 (jQuery). The selected assignment options are implemented with native JavaScript. jQuery syntax and AJAX remain part of the theory preparation.